ZUNFT

The Complete Beginner's Guide

What we have built so far, how the app works, and what comes next - explained for someone who has never worked with any of this technology.

Status: Steps 1 to 6 of the build plan are finished and tested. Step 7 (a real test with real keys and phones, then release) is next.
Written 3 October 2026.

How to read this guide

This guide assumes **zero technical knowledge**. Every technical word is explained the first time it appears, and there is a glossary (Chapter 3) you can flip back to at any time.

You will see four kinds of coloured boxes:

> **IN PLAIN WORDS**
> The simplest possible explanation of the idea.

> **ANALOGY**
> A comparison with everyday life, so the idea sticks.

> **GOOD NEWS**
> Something that works well or was verified.

> **BE CAREFUL**
> A limit, a risk, or something not yet verified. We are honest about these.

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1. The idea

Zunft (German for "guild", the traditional association of craftsmen) is a phone app for small German craft businesses - *Handwerksbetriebe*. The first customer we are designing for is an electrician with 1 to 10 employees in Hamburg.

The problem it solves

Picture Thorsten, an electrician. He spends the day on building sites: installing sockets, fixing fuse boxes. When he gets home at 7 pm he still has to write quotes and invoices - typing numbers into a computer after a long day. Many craftsmen hate this and lose hours (and money) every week. Some forget to invoice at all.

The solution

1. Thorsten opens the job (say "Müller, Keller") and speaks into his phone
"New socket in the basement, two hours, material about 40 euros." The customer is already fixed by the job.

▼

2. The app turns speech into text, then an AI turns text into a structured draft
Work items, hours, material, prices - all separated correctly. Several notes (even over several days) collect on the same job.

▼

3. When the job is done, Thorsten checks all its notes as one draft, fixes anything, taps "PDF"
A professional German invoice, ready to send by WhatsApp or e-mail.

Total effort: about one minute instead of twenty.

Why German craft businesses specifically?

- **German invoice rules are strict.** An invoice needs specific information (address, tax number, invoice number, and so on). The app knows these rules so the craftsman does not have to.
- **A tax rule called § 35a EStG.** Private customers can deduct part of a craftsman's bill from their income tax - but *only the labour part, not the material*. So the invoice must show labour and material separately. Done wrong, the customer loses money and complains. The app separates them automatically.
- **Craftsmen work where there is bad network** (basements, new buildings). So the app must work *without internet* and catch up later.

The long-term vision (three layers)

Layer	What	Status
1. The hook	Voice note to quote/invoice PDF (what we are building now, "Phase 1").	In progress
2. Overflow network	When a business is fully booked it can pass jobs to nearby colleagues, with a referral fee ("Phase 3").	Later
3. Compliance	Legally tamper-proof records (GoBD) and electronic invoice formats (ZUGFeRD / XRechnung) ("Phase 4").	Later

IN PLAIN WORDS

We are only finishing Layer 1. Layers 2 and 3 are deliberately postponed. The database already has a few small "hooks" for them so we do not have to rebuild later.

2. The big picture

The whole system has **three pieces**. Understanding these three is 80% of understanding the project.

PIECE 1 - The phone app (Flutter)
What Thorsten holds in his hand. Screens, buttons, the microphone, a small private database on the phone. Lives in the folder `zunft_app/`.

▲▼ talks over the internet ▲▼

PIECE 2 - The cloud (Supabase)
A rented online "back room": the main database, the login system, and file storage for recordings and PDFs. Lives in the folder `supabase/`.

▲▼ asks for help ▲▼

PIECE 3a - Deepgram
Specialist company that turns **speech into written text** (German).

PIECE 3b - Google Gemini
AI that **reads that text and understands it**: who is the customer, what work, what price.

ANALOGY: A RESTAURANT

The **phone app is the dining room** - what the guest sees and touches. **Supabase is the kitchen and pantry** - it stores everything and does the work guests should not see. The **"Edge Function"** (explained later) is the **head chef** who takes an order and calls two specialists: Deepgram is the *listener* who writes down exactly what was said, Gemini is the *clerk* who understands the note and fills out the form correctly.

Why not put everything on the phone?

- **Secret keys.** Deepgram and Gemini charge money per use. Their passwords (called *API keys*) must never be stored in the app, because anyone can pull them out of an app. They live only in the cloud piece.
- **Shared data.** If the business has several employees, they all need to see the same customers and jobs. That needs one central copy (the cloud).
- **Backup.** Phones get lost and broken. The cloud copy survives.

Why not put everything in the cloud?

Because job sites have bad reception. So the phone keeps its **own small copy** of the data and works with it. When the internet is there, the two copies are synchronised ("sync"). This is called an **offline-first** app, and it shapes a lot of the work in Step 4.

The journey of one voice note

1. **Thorsten chooses the job and records.** The phone saves an audio file (together with the job) and uploads it to Supabase storage in a folder named after his business.
2. **The app calls the Edge Function** ("please process this recording and store it on this job") and shows its login proof.
3. **The Edge Function checks who is asking**, finds his business, and downloads the audio.
4. **Deepgram** turns the audio into German text.
5. **Gemini** reads the text and the business's usual hourly rate (no customer data at all) and returns structured positions.
6. **The Edge Function double-checks the AI's answer** (AI can make mistakes), calculates the tax split and totals, and saves the result as a **note on the job** (with who recorded it and when).
7. **The phone shows the note on the job.** When the work is done, all notes of the job become one draft; Thorsten corrects anything, then issues the invoice and the PDF.

> **WHERE WE ARE**
>
> Every stage of this journey now exists in code, including the last one: a checked draft becomes a numbered, locked invoice with a proper PDF. What has **not** happened yet is one real end-to-end run with real keys on a real phone - see Chapter 11.

3. Glossary

Flip back here whenever a word is unfamiliar.

Word	Plain-English meaning
App	A program on your phone.
Flutter	A toolkit from Google for building one app that runs on **both Android and iPhone** (and Windows) from the same code. Saves building everything twice.
Dart	The programming language Flutter apps are written in.
Code / source code	The written instructions that make a program do things.
Database	An organised electronic filing cabinet.
Table	One drawer of that cabinet, like a spreadsheet: *customers*, *jobs*, *documents*. Each row is one customer; each column is one detail (name, phone...).
Postgres / SQL	Postgres is the database software Supabase uses. SQL is the language you use to talk to it.
Supabase	A company that rents out a ready-made "back room": database + login system + file storage. Saves us from building our own servers.
Server / cloud	A computer somewhere else that is always on and reachable over the internet.
Backend / frontend	Frontend = what you see (the app). Backend = the hidden part behind it (the cloud).
API	A "menu" a service offers to other programs: "send me audio, I send back text".
API key	A long secret password that identifies you to a service and lets them bill you. **Must never be public.**
Edge Function	A small program that lives in the cloud and runs only when asked. Ours is called `process-capture`. It is the "head chef".
AI model / LLM	A program (here: Google Gemini) that can read text and produce text or structured answers. "Low-cost Gemini" means one of their cheaper, faster versions.
Speech-to-text (STT)	Turning a recording into written words. Deepgram does this.
Authentication (login)	Proving who you are (e-mail + password).
Login token (JWT)	A digital "wristband" you receive after logging in. The app shows it with every request so the server knows who is asking.

RLS (row-level security)	Locks *inside* the database: even if two businesses use the same drawer, each can only see its own rows. Like a hotel key card that opens only your room.
Migration	A saved list of changes to the database layout ("add this table, add this lock"). Keeps every copy of the database identical.
Drift / SQLite	SQLite is a tiny database that lives inside the phone. Drift is the helper that makes it pleasant to use from Flutter.
Offline-first	The app works from the phone's own copy of the data and syncs later, so a missing network never blocks work.
Sync	Making the phone's copy and the cloud copy agree: send local changes up, bring remote changes down.
Outbox	A "to-send" tray on the phone. Every change is dropped in it and removed only once the cloud confirms receipt.
Transaction	A group of database steps that either **all** happen or **none** do - like paying at a till: the money and the receipt together, never one without the other. Invoice numbers rely on this.
Embedded font	A typeface stored *inside* the PDF, so letters like ä ö ü ß and the euro sign look the same on every device.
Queue	A waiting line. Voice notes wait in a queue on the phone until they can be processed, one after the other, oldest first.
Idempotent	"Safe to repeat". If the same request arrives twice, the result is the same as if it had arrived once. Every voice note carries a unique number so the cloud never processes it twice.
Riverpod	A Flutter helper that hands the right data to the right screen and refreshes screens when data changes.
Widget	One building block of a screen: a button, a text, a list.
Test	A small program that checks another program behaves correctly. We wrote 351. Like a checklist an inspector ticks automatically.
Git	A "save point" system for code - lets you go back and see who changed what. Set up but nothing saved into it yet.
§ 35a EStG	The German tax law paragraph that lets private customers deduct craftsman *labour* costs (not material) from their taxes.
GoBD	German rules for keeping business records tamper-proof. Finished invoices must not be silently changed.
DIN 5008	The German standard for how a business letter or invoice is laid out.

Cents	We store every price as a whole number of cents (65,00 EUR is stored as 6500). Computers make tiny errors with decimals like 0.1 + 0.2, so money never uses decimals internally.

4. Where we started

When I first read your folder, it already held a detailed plan (`ZUNFT_BUILD_PLAN.md`) and a good amount of code. Think of it as a house with the blueprint and the walls up, but no electricity or plumbing connected.

What already existed and was good

- A **well thought-out plan** with a product description, sprint schedule and a "definition of done".
- A **database design** (six tables: businesses, staff profiles, customers, jobs, capture logs, documents) using whole cents for money.
- A **draft cloud function** for the voice-note processing.
- A **good-looking demo** of the app: a record button, a review screen where you edit line items, and a PDF preview in German invoice layout.
- Correct money calculations with **a few tests**.

What was missing or broken

Problem	Why it mattered
The app was not connected to anything	Everything on screen was made-up demo data. No login, no saving, no cloud.
The record button was fake	It toggled a colour and counted nothing. No audio was ever recorded.
The cloud function had no security	Anyone who found its address could read other businesses' data and run up your AI bill.
Weak database locks	Staff could change each other's roles; invoice numbers could clash or repeat.
No phone-specific folders	The project could only be tested on a PC, not built for Android or iPhone.
Wrong tax handling of flat fees	A call-out fee was treated as material, which would cost customers tax savings.
Invoice details hard-wired	Same invoice number, company and address on every PDF. **Fixed in Step 6.**
Letters like ä ö ü ß in the PDF	The PDF used a font that cannot draw them. **Fixed in Step 6.**
No history, README or setup notes	A stranger (or you in six months) could not run or understand it.

5. Decisions we made together

Question	Decision	Reason
Which AI?	**Google Gemini** (low-cost "Flash-Lite" tier to begin with) instead of Claude.	You asked for low cost. The model name is a setting (`LLM_MODEL`), so it can be switched later without rewriting anything.
Which phones?	**Android and iPhone**	Flutter handles both. Note: building the iPhone version requires a Mac computer (Apple's rule).
Is a call-out fee (Anfahrtspauschale) "labour" for § 35a?	**Yes - counts as labour.** Only material is excluded.	You have no tax adviser, so I decided from my knowledge of the official guidance that travel costs are part of the labour cost. **PLEASE VERIFY** This is from memory and not tax advice. Check the current BMF letter on § 35a before sending real invoices.
Speech-to-text vendor	Keep **Deepgram** for now.	The plan benchmarks it for German trade words. Gemini can also listen to audio directly, which would remove a vendor - we will compare them with real recordings.

6. What was built, step by step

The plan had seven steps. Six are complete. Each section below says **what** was done, **why**, and **how it works**.

Step 1 - Foundation done

Goal: make the project a proper, buildable, documented project.

- **Git created.** The folder is now a Git project, with a `.gitignore` list that keeps junk (build output) and **secret keys** out of the history. *Nothing has been "committed" (saved) yet - I only commit when you ask.*
- **Phone folders added** (`android/` , `ios/` , `windows/`) by running Flutter's own scaffolding command.
- **A real bug fixed:** the colour theme used a name that the current Flutter no longer accepts, so the app did not compile. Fixed.
- **README and `.env.example`** written: how to start the project and which secret settings are needed (with the safe rule: secrets only ever in the cloud, never in the app).
- **Internet permission added for Android.** Android's default template only allows internet access in *test* builds. A real release build would have been unable to reach the cloud at all.

Step 2 - Backend hardening done

Goal: make the cloud side safe and correct. Two parts: the Edge Function, and the database.

2A. THE EDGE FUNCTION `PROCESS-CAPTURE`

> **ANALOGY: THE HEAD CHEF**
> Before, the chef took orders from anyone who shouted through the kitchen door, and trusted whatever the customer said about who they were. Now there is a **doorman**: he checks the wristband (login token), looks up which business you belong to *himself*, and only lets you into your own pantry.

Improvement	What it means
Login required	No valid login token: refused (error 401). The business is found from the user's profile, **not** from anything the caller claims.
Acts as the user	It no longer uses the all-powerful "master key". Every database action runs with the user's own limited rights, so the database locks (RLS) protect each business.
Folder rule for recordings	A recording must live under `<business id>/...`. Nobody can ask it to transcribe another business's audio.
Size limits	Audio max 10 MB, text max 4,000 characters - protects against runaway bills.

Switched to Gemini	Asks Gemini for an answer in a strict JSON shape. The model name is a setting.
Never trust the AI blindly	A checker (`_shared/draft.ts`) rejects nonsense: unknown item types, negative quantities, absurd prices, or a customer ID the AI invented. If the answer is bad it retries once, then returns a friendly error.
Prompt-injection defence	The spoken text could contain "ignore your rules and..." (someone could say it on purpose). The text is wrapped in labelled tags and the AI is told it is only *data*, never instructions.
No leaking	Errors shown to the app are short and generic; technical details go to the server log only.
Bugs fixed	Empty speech result no longer crashes it; audio type is detected from the file; failing to save a job now stops with an error instead of silently continuing.

2B. THE DATABASE MIGRATION

A **migration** is a saved set of database changes. We kept the original and added a second one, `20261002000000_harden_schema.sql`.

Improvement	What it means
Safe invoice numbers	Numbers look like `RE-2026-0001` , `RE-2026-0002` ... The old method ("count existing + 1") could give two invoices the same number if two people saved at once. The new method keeps a **counter per business, type and year** that is locked while it is being increased - like a single ticket machine with one person at a time. If a save fails, the number is handed back, so there are **no gaps** (German tax law dislikes gaps).
Better locks (RLS)	Old: one broad rule per table. New: separate rules for reading, adding, changing and deleting. A job cannot point to another business's customer. Only an **owner** can change roles; nobody can promote themselves. The audit log can be added to but never edited or deleted.
Tamper-proof invoices (GoBD)	Once an invoice is "locked" it can no longer be edited or deleted, and the lock time is stamped automatically. The invoice number can never be changed.
Onboarding function	A new function `create_business_for_user()` creates the business *and* its owner in one go (both or neither). Apps cannot insert into those tables directly.
Invoices for brand-new customers	A document can now store a **copy of the recipient** (name and address) instead of needing a customer record. That also freezes what was printed even if the customer is edited later.
Sanity checks	Allowed statuses/roles are enforced; totals must add up (net + VAT = gross); "last changed" times update automatically.
Safe helper functions	The internal helpers were hardened and cannot be called by logged-out visitors.
File storage rules	Recordings and PDFs are private; each business can only touch files in its own folder.

Step 3 - Login and business setup done

Goal: a real front door for the app.

App starts
Is the cloud address configured? If not, it shows a short "how to configure" message.

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Not logged in?
→ Login / Register screen

▼

Logged in but no business yet?
→ "Set up your business" (two questions: business name, your name, plus trade)

▼

Everything set?
→ The main app with three tabs

This decision logic lives in one small piece called the **AuthGate** (the "bouncer"). It reacts live: log out and you are instantly back at the login screen.

- **Login screen:** e-mail and password, show/hide password, clear German error messages ("E-Mail oder Passwort ist falsch"), password at least 8 characters on sign-up, and a friendly note if e-mail confirmation is switched on.
- **Business details screen ("Betrieb" tab):** the info printed on every invoice - address, tax number, VAT ID, bank details, standard hourly rate and VAT rate. It checks input as you type: a real **IBAN checksum test**, VAT-ID format, sensible rates. It accepts German commas ("68,50").
- **Missing-information warning:** a yellow banner lists what an invoice still lacks (address, tax number/VAT ID, IBAN).
- **Roles:** only the owner and the office role can edit business data; others can view. (The database enforces this too - the app is just being polite.)
- **Price parser fixed:** typing "1.5" used to be understood as 15 cents. It now understands both German ("1.250,50") and plain ("65.50") ways of writing prices, and rejects nonsense.

Step 4 - Local database and sync done

Goal: real customers and jobs, working **without internet**.

ANALOGY: A NOTEBOOK AND A FILING CLERK

Thorsten carries a **notebook** (the phone's local database). He writes everything in it immediately, wherever he is. Every new note also goes into a **"to-send" tray** (the outbox). Whenever there is internet, a **filing clerk** (the sync engine) carries the tray's contents to the head office, and brings back any new papers from there. A note leaves the tray only after the office confirms it arrived.

WHAT YOU SEE

- **Aufträge (Jobs) tab:** list of jobs with customer and coloured status; a summary of drafts / in progress / completed; add, edit, delete.
- **Kunden (Customers) tab:** list with search (name, street, city, phone - several words allowed), add, edit, delete (asks first).
- **Sync tile in the Betrieb tab:** honest status - "Synchronisiert", "Offline", "Please sign in again", or "Some changes were rejected", with the last sync time and how many changes are waiting. Tap it to sync now.

HOW THE SYNC WORKS, IN A WORKED EXAMPLE

1. In a basement with no signal, Thorsten adds the customer "Schmidt". It appears in the list **immediately** and a note "upload customer Schmidt" lands in the outbox.
2. He then adds a job for Schmidt. A second note lands in the outbox.
3. Back in his van the phone gets signal. The sync engine wakes up. **Order matters:** customers are sent *before* jobs (a job pointing at a customer the cloud has not seen would be refused). Deletes go the other way round: jobs first, then customers.
4. Each note leaves the outbox only when the cloud confirms it. If the cloud *refuses* one (e.g. a rule is violated) that note stays, with the reason, and the others continue. If the connection drops, nothing is lost - it retries later.
5. Then the engine **pulls**: it asks the cloud "what changed since I last asked?" (a bookmark, called a *watermark*, remembers where it stopped), and also removes local rows that were deleted elsewhere.

WHEN DOES A SYNC HAPPEN?

When the app opens, a couple of seconds after any change, as soon as the network returns, every 5 minutes, and whenever you tap the sync tile. Only one sync runs at a time.

WHAT IF TWO PEOPLE EDIT THE SAME THING?

The rule is **"last write wins"**: whoever syncs last decides. A change still waiting in the outbox is *never* overwritten by incoming data. This is simple and fine for a small team, but two people editing the *same* customer at the same moment could overwrite each other.

TWO BUGS FOUND AND FIXED DURING THE BUILD

LOST EDIT

If you edited a record *while* the previous version was being uploaded, the app could have deleted your newer edit from the outbox when the older upload finished. Each outbox entry now carries a revision number, and it is only removed if that number is unchanged. A test proves it - and I confirmed the test fails when the protection is turned off.

ACCIDENTAL DELETION AT SCALE

The cloud returns at most 1,000 rows at a time. The "remove what was deleted elsewhere" step would have mistaken customer number 1,001+ for deleted ones. The list is now fetched page by page.

SHARED PHONE SAFETY

The phone remembers which business its cached data belongs to. If a *different* business logs in on the same phone, the old cache is wiped first so nobody sees someone else's customers.

TOOLING NOTE

Drift needs a "code generator" that writes helper code from the table definitions. It crashed at first because its internals were older than your Flutter version. I upgraded it and removed a second generator (`riverpod_generator`) that nothing used. The generated file is saved in the project so you do not need to run it again unless the tables change.

Step 5 - The real capture flow done

Goal: the "magic" step: speak (or type) a note, and get a draft - even with bad reception.

WHAT YOU SEE

1. Tap the amber **SPRECHEN** button, choose the job (see Step 6b), and a sheet opens with a big microphone button. The first time, the phone asks for permission to use the microphone.
2. Tap to start. A timer counts up. Tap again to stop. (The limit is **3 minutes**: the cloud only reads the first 4,000 characters of the text, so longer notes would be cut off silently. 3 minutes of speech stays safely below that.)
3. The sheet closes at once with "Aufnahme gespeichert". You do **not** wait. While it is not yet processed it appears at the top of the Aufträge tab under **WARTEN AUF VERARBEITUNG**.
4. A moment later it disappears from that list and shows up **on the job**, with its positions and the AI's warnings, for example "price estimated". (Since Step 6b a processed note lives on the job in the cloud, not in the phone's queue.)

No microphone, or too noisy? The same sheet has **"Lieber tippen"** (type instead): the typed text goes through exactly the same pipeline, minus the speech recognition.

THE LIFE OF A NOTE

Status shown	Meaning	What happens next
Waiting (amber)	Saved on the phone, not yet processed. If there is no internet it says "Wartet auf Internet".	Sent automatically as soon as possible. Nothing for you to do.
Processing (blue)	Being uploaded and turned into a draft.	Usually seconds.
Done	The cloud stored it as a note on the job.	It leaves this list and appears on the job screen.
Failed (red)	Shows the reason in plain German.	**Wiederholen** (try again) or **Verwerfen** (discard).

> **ANALOGY: A PARCEL WITH A TRACKING NUMBER**
>
> The voice note is a parcel. The phone puts it in an **outgoing mailbox** (the queue). A **postman** (the processor) delivers it whenever the roads are open. Every parcel has a unique **tracking number**. If the postman delivers the parcel but the "delivered" receipt gets lost, he tries again next time - and head office says "we already have that one, here is the result again" instead of storing the note twice and paying the AI twice.

BUILT TO SURVIVE INTERRUPTIONS

What goes wrong	What the app does
No internet when you stop recording	The note waits in the queue and is sent automatically when the network returns.
Signal drops *after* the audio was uploaded	The retry does **not** upload the audio again and uses the same tracking number, so no duplicate job appears.
The cloud has a temporary hiccup	Up to 3 automatic tries, then "failed" with a clear message and a retry button.
The AI could not make a usable draft	Fails straight away with the server's message (retrying the identical input rarely helps). You can try again or type it.
The app is closed or killed mid-processing	At the next start, unfinished notes are put back in the queue.
Login expired	Everything pauses and nothing is lost; the note waits until you sign in again.
You tap the mic by accident (under 1 second)	"Die Aufnahme war zu kurz" - nothing is queued.
You close the sheet while recording	The recording is cancelled, so the microphone is never left running.

The audio file is deleted from the phone only **after** the cloud confirmed the note.

TECHNICAL DETAILS, IN PLAIN WORDS

- **Recording format:** speech is recorded mono, 16 kHz, AAC (a small compressed format): about half a megabyte per minute. Noise suppression and automatic volume are switched on for building sites.
- **Where the audio goes:** into private cloud storage, in a folder named after the business (`<business id>/<note id>.m4a`), matching the cloud function's folder rule.
- **Phone permissions:** Android needs `RECORD_AUDIO` declared; iPhone needs a sentence explaining why the microphone is used (otherwise the app would crash when asking). Both were added.
- **Database change on the phone:** a new table for queued notes. The local database moved from version 1 to 2 with a proper upgrade step, so existing installs keep their data.
- **Cloud change:** a third migration adds a "capture id" column with a uniqueness rule, and the Edge Function now recognises repeats (and handles two identical requests arriving at the same moment by keeping one result).

BUGS THE TESTS CAUGHT WHILE BUILDING THIS

THE MICROPHONE COULD HAVE STAYED ON

If the sheet was dismissed while recording, the cleanup code used a Riverpod feature that is not allowed at that moment. It would have thrown an error and left the microphone recording. The test "closing the sheet while recording" found it; the code now takes what it needs at start-up.

DUPLICATE ITEM IDS

AI drafts have no item IDs, so the app invented them from the clock, in milliseconds. Several items created in the same millisecond got the *same* ID, which would have broken editing and deleting. Each item now gets a unique ID.

FILE ACCESS COULD NOT BE TESTED

Reading the audio file straight from disk meant the processing never finished inside a test (real disk access cannot run inside the test simulator's fake clock). File access now sits behind a small interface, like the microphone and the cloud, so each can be swapped for a pretend version in tests.

Step 6 - Real documents: finalize, number, PDF done

Goal: turn a checked draft into a legally usable quote or invoice: a gapless number, a frozen record, and a PDF you can send.

WHAT YOU SEE

1. **Review the draft.** Open a job and tap **Alle Notizen prüfen und abrechnen**: all the job's notes appear as one draft. Fix anything: the customer's name and address, each position (type: Arbeitslohn / Pauschale / Material, quantity with a comma like "1,5", price), add or delete positions. A yellow box lists the AI's warnings. The VAT line follows your business setting (19 %, 7 %, or 0 % for small businesses).
2. **Look at a preview** ("Vorschau als PDF"). It is clearly stamped **Entwurf** in red, shows "wird vergeben" instead of a number, and has no share button - so a draft can never be sent out by mistake.
3. **Tap "Rechnung fertigstellen".** A last-look box asks for the *Leistungszeitraum* (when the work was done), the payment term (7, 14 or 30 days; for quotes: valid 14, 30 or 60 days), whether to add a new customer to your list, and, if any price is still flagged "Preis prüfen", that you have checked it.
4. **The number is issued** (for example RE-2026-0001), the document is saved, the PDF is made and a copy is stored online. The PDF opens with **share** (WhatsApp, e-mail) and **print**.
5. **Tidy-up happens by itself:** the job's notes are used up (they will not appear on a next invoice), the job moves to "Abgerechnet" (or "Angebot versendet"), and, if the job had no customer yet, the new customer is created - with the address split into street, postcode and city - and linked to the job.

WHY YOU NEED INTERNET TO FINALIZE

> **ANALOGY: THE TICKET MACHINE AT HEAD OFFICE**
>
> Draft documents can be edited anywhere, even in a basement. But invoice numbers must be consecutive with **no gaps** (German tax auditors check this). Only one place can guarantee that: a single ticket machine at head office (the cloud). So the final step needs a connection. If it fails, the app says so, **keeps all your edits**, and you simply try again later.

HOW THE NUMBERS ARE TRULY GAPLESS

In Step 2 the number was handed out by one cloud call and the document was saved by a *second* call. If the connection died in between, a number would be used up with no document behind it: a gap. Now **one cloud function (`create_document`) does everything in a single transaction**: check the input, calculate the totals, take the next number, save the document. Either all of it happens or none of it does. Retrying the same request (same document ID) returns the same document and the same number, never a second one.

WHAT THE CLOUD CHECKS BEFORE IT ISSUES ANYTHING

Check	Why
Totals are calculated by the cloud, never taken from the phone	The phone cannot send wrong or tampered amounts. The VAT rate also comes from the business settings in the cloud.

An invoice needs the sender's full address and a tax number or USt-IdNr.	Required on every invoice by § 14 UStG. (The app checks the same thing first, so you get a clear message without waiting.)
Over 250 € gross, the customer's address is required	Small invoices up to 250 € may leave it out (§ 33 UStDV); larger ones may not.
Every position is valid: type, unit, text, quantity, price	No empty, negative or absurd positions. (Missing fields are caught too; see the bugs below.)
Customer and job must belong to your business	No mixing up with another business's records.
Dates are sensible	No invoice dated years ago or in the far future.
Documents cannot be inserted any other way	The old direct "insert" permission was removed; only this one function can create a document.

AFTER IT IS ISSUED

The content is **frozen** (GoBD): amounts, positions, recipient, number, dates, even the lock itself can never change, and it can never be deleted. Only two things may still change: the **status** (Erstellt → Versendet → Bezahlt, never backwards) and the PDF file path. For corrections you create a new document.

THE PDF, BEFORE AND AFTER

Before	Now
Letters like ä ö ü ß and € could drop out; spellings like "fuer" were a workaround	An embedded font (Lato, open licence): all German letters, € and § print correctly, with proper wording
Same invoice number, sample company and sample address on every PDF	The real number, your business data, the customer from the document
"19 % MwSt." fixed text	The real rate; with 0 % the line is replaced by "Gemäß § 19 UStG wird keine Umsatzsteuer berechnet."
Leistungszeitraum always the issue date	What you entered; due date or validity date included
Always one page	Long documents flow onto more pages, with "Seite 1 von 2" and the bank details in the footer
§ 35a note showed the net labour amount	Shows labour **including VAT** (the amount the deduction is based on), invoices only

I LOOKED AT THE RESULT

I did not just check that the PDF "exists". I rendered an invoice, a quote, a draft preview and a small-business invoice to images and looked at them: umlauts, ß, €, §, the numbers (for example 102,00 + 80,00 = 182,00 net, 34,58 VAT) and the layout were correct.

THE NEW BELEGE TAB

- **List** of all quotes and invoices, newest first, with colour-coded status and a summary: number of open invoices and the open amount.
- **Detail screen:** what the document says, status, and **"PDF ansehen und senden"**. It fetches the *original* PDF from storage; if that is not possible it recreates the PDF from the stored data.
- **Status buttons:** "Als versendet markieren" and (invoices only) "Als bezahlt markieren". Sharing the PDF also marks it as sent. Status changes need internet because the cloud keeps the master copy.
- Documents are kept up to date on the phone by the same sync as customers and jobs (read-only: they are only ever created by the cloud function).

MISTAKES FOUND AND FIXED WHILE BUILDING THIS

A HOLE IN MY OWN VALIDATION

In database language, "missing" is not the same as "wrong": a check like *type NOT IN (...)* silently passes when the type is missing altogether. A request with no positions at all could have produced an empty invoice. Every field is now checked for "missing" explicitly, and the tests try each missing field one by one and confirm that nothing is stored and no number is used up.

A MISTAKE FROM STEP 2

I had frozen locked documents so completely that nobody could mark an invoice "sent" or "paid". The rule now freezes the content but still allows the status workflow (migration 4 fixes this).

DUE DATES OFF BY A DAY

Computing "today + 30 days" as 30 × 24 hours is wrong when the clocks change (last Sunday of October): the due date lands one day early. Now it counts calendar days. A test crossing the clock change caught it.

A SPINNER BEHIND THE ERROR MESSAGE

When finalizing failed, the progress spinner kept turning behind the error box until you closed it. The busy state is now reset before any message is shown.

A NEWLY ISSUED DOCUMENT COULD VANISH FOR A MOMENT

Sync removes local documents that the server no longer lists. If you issued a document *while* a sync was running, that list was out of date and the new document was removed locally (it came back at the next sync). Sync now never removes a document that is newer than what it has already seen.

A QUIRK OF THE DATABASE LANGUAGE

A *CASE ... THEN* inside an *IF ... THEN* condition is cut off at the first THEN. Found by the tests; the value is now computed first.

Step 6b - The redesign: one job, many notes done

Why: you asked good questions after Step 6, and they exposed a weakness. In the first version *every voice note became its own draft*, and the AI *guessed the customer* from the spoken name. With many people called Müller that is dangerous, and a job spread over several days would have produced a pile of small drafts. So the working of the app was changed, before any real customer used it.

THE NEW IDEA IN ONE PICTURE

1. Choose the job first

Tap SPRECHEN, pick "Schiller, Badezimmer" (or "Neuer Auftrag"). The job already knows its customer.

▼

2. Record as many notes as you like - today, tomorrow, by any worker

A big banner "AUFNAHME FÜR: Schiller / Badezimmer" shows where the note goes. Wrong job? Tap "Ändern".

▼

3. The AI only turns each note into positions

Hours, material, flat fees. It does not choose or look up a customer, and it is not even sent your customer list.

▼

4. When the job is done: "Alle Notizen prüfen und abrechnen"

All notes of the job become ONE draft. Check it, then issue ONE numbered invoice.

WHAT CHANGED FOR YOU

Before	Now
SPRECHEN started recording straight away	It first asks **"Für welchen Auftrag?"** (the job). Every note belongs to exactly one job.

The AI matched the spoken name to a customer	The **customer comes from the job**, which you chose. If the speaker names someone else, nothing changes. Only when a job has *no* customer, the spoken name is *offered* as a suggestion you must tap.
Each note was a separate draft	A job shows **all its notes** (who recorded them, when, the positions, what was said). One button turns them into one document.
Tapping a job opened its edit form	It opens the **job screen**: customer, "Notiz aufnehmen" button, notes, total, and the billing button. Editing is the pencil icon.
A wrong note could only be edited on the review screen	Each note has a bin icon: it is removed from the job (it stays in the audit log, but is never billed).

SEVERAL WORKERS, ONE JOB

The notes are stored in the **cloud**, on the job, with the name of the worker who recorded them. So if Thorsten records on Monday and Gesa on Tuesday, both see both notes, and one of them issues **one** invoice. The record shows who did what and when; this is the beginning of the "worker follow-up" you asked about. (A full time-tracking view for the boss is still a later project.)

NO DOUBLE BILLING

> **THE NOTES ARE "USED UP" BY THE INVOICE**
>
> When the invoice is issued, the cloud marks the notes it contained as used, *in the same single step* that issues the number. They can never appear on a second invoice. Notes recorded after the invoice start the next document. If the connection drops during issuing, the retry returns the same invoice and does not touch other notes.

NOTES RECORDED WITHOUT INTERNET

Unchanged in spirit, with one new detail. A note is saved on the phone together with its job. If the *job itself* was also created offline, the app first sends the job and then the note, because a note can only be stored on a job the cloud already knows. Until then the note shows "Der Auftrag wird noch synchronisiert". Once the cloud has stored a note, the phone's copy is removed, since the cloud is now the single source of truth. The job screen warns "x weitere Notiz(en) warten noch auf die Verarbeitung" if some are not yet included.

TECHNICAL DETAILS, IN PLAIN WORDS

- **Fifth migration:** notes get an author, a "used in document" link and a "removed" flag. A note cannot be attributed to a colleague or created as already used. Removing a note is a small guarded cloud function (only inside your own business, only if not yet billed).
- **The cloud function** now *requires* a job and no longer creates jobs. It reads no customer data and its prompt contains no customer list. It returns only a name *hint*.
- **The invoice-making function** (`create_document`) accepts the note IDs and uses them up atomically.

- **Privacy effect:** the AI no longer receives customer names, first names or cities from your database. It only gets what is spoken.
- **Tests:** 9 new database tests (two workers, no double billing, foreign notes untouched), the cloud-function tests reworked (now 11), and a rewritten app test set (213, including the duplicate check): picking a job, banner, "Ändern", offline notes for several jobs, job created offline, combining notes from two workers into one invoice, a job without customer, notes used up.

DUPLICATE CUSTOMERS: A NAME ALONE IS NOT ENOUGH

Many people are called Müller, so the app does **not** treat a matching name as "same customer". When you create a customer by hand it compares more than the name:

Situation	What the app does
Same name and same address (street, number, postcode), or same phone, or same e-mail	Clear warning "Diesen Kunden gibt es schon". You can cancel or create it anyway.
Same name, but a different address	No warning at all: it is a different person.
Same name, but the address (or phone/e-mail) is missing on one side	A softer question "Gibt es diesen Kunden schon?". You decide.
Different name, even at the same address	No warning (a family or a firm at one address).

Spelling differences are ignored: "Müller" = "Mueller", "Müller, Thomas" = "Thomas Müller", "Hauptstraße 5" = "Hauptstr. 5", phone numbers in any format. House numbers do matter (no. 5 is not no. 7).

When an invoice is issued for a customer typed in on the review screen, the app only **reuses** an existing customer in the clear case (same name **and** same address/phone/e-mail). In the unclear case it keeps them separate, because a wrong automatic merge would put an invoice on the wrong person's record, which is worse than a duplicate.

A BUG THE TESTS CAUGHT

> **INVISIBLE TAP EFFECT**
> The job picker drew its list on a plain coloured box. Flutter warns that a list row's tap highlight would be hidden by it. The picker now sits on a proper material surface.

STILL OPEN HERE

- Duplicate customers are now **warned about** (see below), but there is still no "merge two customers" function for duplicates that already exist.
- A note's **audio** is kept in the cloud but cannot be played in the app yet (the written transcript can be shown with "Was wurde gesagt?").
- Interim invoices (Abschlagsrechnung) are not special: you simply issue a document from the notes so far and keep recording.

- Who sees which jobs is now solved by roles and job assignment (see Step 8).

Step 8 - Teams: roles, assigned jobs, invitations, seats done

Goal: make Zunft work for a self-employed craftsman *and* for a boss with employees, in the **same app**, so it can be offered to any craft business in the app stores. One idea carries all of it: **one kind of account; a business has a number of seats and people have roles.** A self-employed person is simply a business with one seat. If they grow, nothing is converted: the plan gets more seats and all jobs, customers and invoices stay.

WHO CAN DO WHAT

Role	Sees	Can
Inhaber (owner; also the self-employed person)	Everything.	Everything, plus: invite people, switch people off, see the plan.
Büro (office)	Everything.	Assign jobs, issue and see invoices, edit business data. Cannot manage the team.
Meister, **Geselle** (employees)	Only the jobs **assigned to them**, those jobs' customers and notes, and the customers they created themselves.	Record notes, create jobs and customers in the field. **No** invoices, no Belege tab, no other jobs.

ENFORCED IN THE DATABASE, NOT JUST HIDDEN

The rules above are "row level security" in the cloud. Even a hacked phone cannot read another job, another worker's recording or any invoice: the cloud simply does not hand the data over. Screens only adapt to match.

HOW A BOSS ADDS AN EMPLOYEE

1. **Boss:** Betrieb → **Team** → **Mitarbeiter einladen**, choose Geselle, Meister or Büro. The app shows a **QR code** and an **8-character code**. The code works once and expires after 7 days. It already reserves a seat.
2. **Employee:** installs Zunft and **registers with their own e-mail and password**. (Nobody ever shares the boss's login, so you can always see who did what.)
3. **Employee:** on the "Betrieb einrichten" screen taps **"Mit Einladungscode beitreten"**, enters their name and **scans the QR code** (or types the code). They are in, with the invited role.
4. **Boss:** opens a job, taps **Zuweisen** and ticks who works on it. The employee's phone then shows that job (and its customer) after the next sync.

If someone leaves, the owner taps **Abschalten**: access ends immediately, the seat is freed, **their notes stay** on the jobs. The person sees "Dein Zugang wurde abgeschaltet" and nothing of the business. They can be switched on again.

SEATS AND PLANS

- Every business has a **seat limit** (new businesses start with 1). Active people *and* open invitations count. The cloud refuses a join or an invitation when no seat is free, so the limit cannot be bypassed by editing the app.
- The Team screen shows "x von y Plätzen belegt".
- **The limit can only be changed on our side** (not from the app). **Payments are not built**: there is no purchase screen yet. For testing you raise the limit by hand (Chapter 15, Part E).
- A self-employed person sees none of the team features: no assignment card, no "Zuweisen" (nobody to assign to).

LEAVING, REMOVAL AND DELETING AN ACCOUNT

You made the key point: **the account belongs to the person** (they made it with their own e-mail), the business only has a *link* to it. So the boss can end the link, but never lock someone out of their own account. Three actions:

Action	What happens
Abschalten (owner, temporary)	Access is paused, the seat is freed, they can be switched on again. The person sees "Dein Zugang wurde abgeschaltet".
Aus dem Betrieb entfernen (owner, final)	Every link ends at once: no jobs, no customers, no data, off all assignments. The person **keeps their account** and is simply "without a business" again: they can **start their own business** (for example go solo on their own plan) or **join another one** with an invitation. Their **notes stay** on the jobs (the business needs its work records), with the author's name saved as text on the note.
Betrieb verlassen (the person, any time)	The same as being removed, started by the employee.
Konto löschen (the person)	Erases the account completely (the app stores require this, and so does the GDPR). Notes stay on the jobs without any link to a live account. **An owner cannot do this with a button**: a business's invoices must be kept for years by law, so closing a business is a deliberate step through support.

- **One business per account** (your decision): a worker with a weekend side business uses a second account with another e-mail for it.
- **The phone forgets.** When access ends, the business's data is wiped from that phone. If the owner removes someone while their app is open, the app notices after its next sync (within minutes, or when the network returns) and switches to the start screen.
- **Shared phones:** the saved data on a phone now belongs to one *person* in one business, not just to the business. If a colleague logs in on the same phone, the previous person's jobs are wiped first.
- **Privacy notice for employees:** before joining, an employee must tick that they read a short notice (what is stored: name, notes with time and author, recordings, assignments; no location, no time tracking; the service providers; their rights; that the account stays theirs). It is also under Betrieb → Datenschutz.

THE PRIVACY NOTICE IS A DRAFT, NOT LEGAL ADVICE

I wrote it from how the app really behaves. Before real employees use it, a lawyer or data-protection officer must review it and fill in what only you can (who exactly is responsible, legal bases, retention periods, contact address). The same goes for a privacy policy for customers' data and the data-processing agreements. The "support" contact mentioned for closing an owner's account does not exist yet either.

THINGS I HAD TO THINK ABOUT

Question	Decision
A newly assigned job was invisible on the employee's phone	Phones fetch "what changed since last time". An old job would never count as changed. Assigning now refreshes the job's and its customer's change date, so it shows up.
What can an employee do about customers?	See those of their jobs, and those they created. They do **not** see the whole customer list. Trade-off: the duplicate-customer warning only knows the customers they can see.
Recordings (audio)	Everybody can upload; you can only fetch your own, and the boss and office can fetch all.
Why a code and not a login created by the boss?	Own logins are safer (nobody knows anybody else's password), leaving is a single switch, and it is less work for the boss.
Can one person be in two businesses?	No, by decision: one business per account. A weekend side business gets its own account with its own e-mail. A person who leaves can start their own business or join another with the same account.

TECHNICAL DETAILS, IN PLAIN WORDS

- **Sixth migration:** new tables for assignments and invitations; "active" flag on people; seat limit on the business; new access rules for jobs, customers, notes, documents and files; guarded functions to invite, join, switch people off and assign jobs; invoice creation is now refused for employees.
- **The invitation code** uses letters and digits without look-alikes (no 0/O, 1/I), is single-use and expires. Every failure says the same thing ("invalid invitation"), so codes cannot be probed.
- **In the app:** a Team screen, a join screen (QR scanner and typing), an assignment card on the job, role-dependent tabs, a "switched off" screen. The QR code is drawn directly: the ready-made QR widget crashed inside a dialog (found by a test), the same bug would have hit the boss's phone.
- **Tests:** 22 new database tests (every rule above, both allowed and refused), 32 new app tests.

NOT VERIFIED, NOT BUILT

> **CAMERA SCANNER IS UNTESTED ON A REAL PHONE**
>
> The QR *code display* and all the logic are tested. The live camera scanner (the `mobile_scanner` package) cannot run in an automatic test and could not be tried here, so the first real test must check it. If it misbehaves, typing the 8-character code always works.

- **Payments and plan changes:** not built; app-store rules for selling subscriptions inside an app still need checking.
- **Changing a person's role** later (e.g. Geselle to Büro): not in the screen yet (invite again with the new role, or the owner changes it in the database).
- **Time tracking and a boss overview of hours:** not built; the notes already record who and when.
- **Legal texts:** a draft employee notice exists (see above), but it needs a lawyer's review. Customer-facing privacy policy and processing agreements are still to do.
- **Closing an owner's account** (and what happens to the business's invoices) is a manual support process, not an app function.

7. How the app works

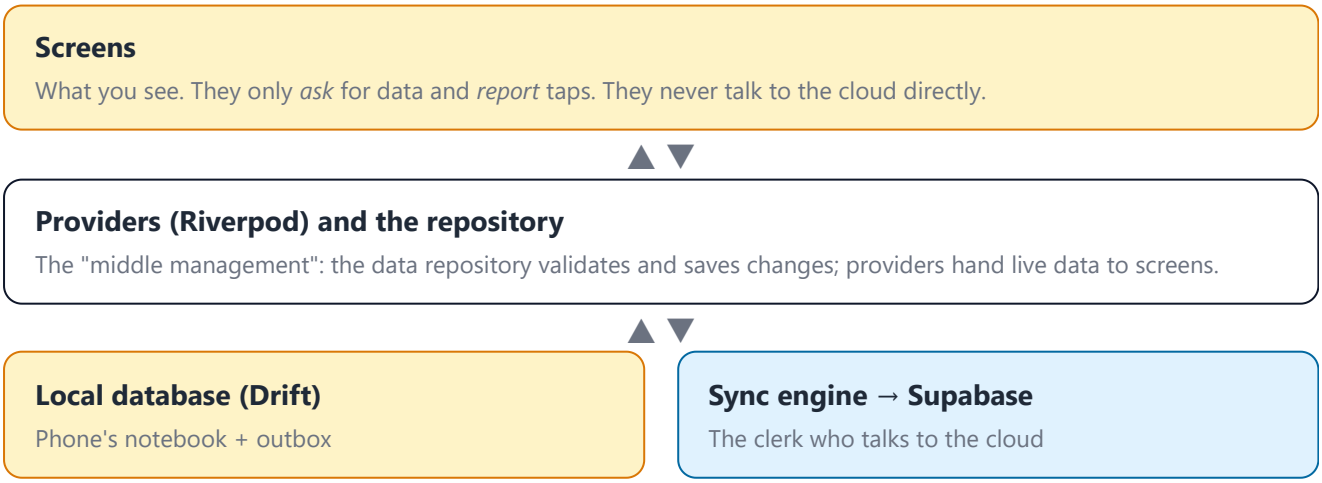
This chapter ties it together: what each part does when you use the app.

The tabs

Tab	Shows	Data comes from
Aufträge	Your jobs, status chips, a small summary - and, at the top, **WARTEN AUF VERARBEITUNG**: notes of this phone that are waiting or failed. Tap a job to open its **job screen**: notes, recording, billing.	Jobs: the phone's local database (kept in sync). Notes of a job: read from the cloud.
Kunden	Your customers, search.	Phone's local database.
Belege ***(boss and office only)***	All issued quotes and invoices with status, open invoices and amount; tap one for details, PDF and status buttons.	Phone's local copy, pulled from the cloud by sync (read-only).
Betrieb	Your business details, sync status, log-out and, for the owner, **Team** (invitations, seats, switching people off).	Business details come from the cloud; sync status from the sync engine.

The big amber **SPRECHEN** (speak) button is always visible. It asks which job the note is for, then opens the recording sheet (Chapter 6, Step 5 and 6b). On the job screen, "Alle Notizen prüfen und abrechnen" opens the review screen, where "fertigstellen" issues the numbered document (Chapter 6, Step 6).

Three layers inside the app



WHY THIS DESIGN?

Screens always read from the **local** notebook, so they are instant and never "wait for the internet". Only the sync engine deals with the network. This is also what made it testable: in tests we swap the real cloud for a pretend one.

What happens when you save a customer

1. You tap **Speichern**. The form checks the name is not empty.
2. The repository trims spaces, turns blank fields into "empty", creates a unique ID (a long random code - created on the phone so it works offline), and writes the customer **and** the outbox note in one all-or-nothing step.
3. The list updates instantly (screens watch the local database).
4. A sync is scheduled a moment later. If the network is there, the customer reaches the cloud; if not, it waits in the outbox.

What happens at login

1. Supabase checks e-mail and password and gives the app a login token (stored safely on the phone, so you stay logged in).
2. The app asks: "does this user have a profile?" If not → business setup. If yes → main app.
3. The app checks the local cache belongs to this business (wipes it otherwise), then starts syncing.

8. The money and tax rules inside the app

Labour vs. material (§ 35a)

Every line on a quote or invoice has a type:

Type	Examples	Counts for the customer's tax deduction?
labor	"2 hours: install socket"	Yes
flat_fee	"Call-out fee"	Yes (our decision - please verify)
material	"Gira double socket, 40 EUR"	No

The invoice shows a note for private customers: the labour amount **including VAT** (the deduction is based on the gross figure), the net amount, the material amount (stated as not eligible), and a reminder to **pay by bank transfer and keep the receipt** - cash payments are not deductible.

A worked example

Line	Type	Net
Installing socket, 2 h x 68,00 EUR	labor	136,00 EUR
Gira double socket	material	40,00 EUR
Net total		**176,00 EUR**
VAT 19 %		33,44 EUR
Gross total		**209,44 EUR**
Labour including VAT (for the tax note)		161,84 EUR

This exact example is used in our tests on **both** the phone side and the cloud side, so the two always agree.

Other rules

- **Money in cents.** 65,00 EUR is stored as 6500 so rounding errors cannot creep in.
- **Invoice numbers** are sequential per business, type and year and gap-free (Chapter 6, Step 6). Quotes use `ANG-`, invoices `RE-`. The number is issued in the same cloud step that saves the document, only when it is finalised, never for drafts.
- **The cloud calculates the final amounts** from the positions; the phone only shows a preview with the same rules.

- **Small businesses (§ 19 UStG):** with a VAT rate of 0 % the invoice carries the mandatory sentence instead of a VAT line.
- **Invoices over 250 €** must show the customer's address; small ones may leave it out.
- **Finished invoices are locked** (GoBD).
- **VAT rate** is a business setting (19 % default).

9. Security

A business app holds customers' names, addresses and money. Here is the protection, from the outside in.

Layer 1 - Login.
No wristband, no entry. Passwords are handled by Supabase, never stored by us.

▼

Layer 2 - The Edge Function doorman.
Works out the business from the profile; limits file sizes; only allows files from the business's own folder.

▼

Layer 3 - Database locks (RLS).
Even with a valid login, you only see and change rows of your own business. Cross-business links are refused.

▼

Layer 4 - Rules inside the database.
Roles cannot be self-promoted; documents can only be created through one controlled function; finished invoices are frozen; the audit log is append-only.

THE GOLDEN RULE ABOUT SECRETS

Only two kinds of keys exist: the **public** key (safe inside the app, it only allows what the locks permit) and the **secret** keys (Gemini, Deepgram, the master key). Secret keys live *only* in the cloud. The `.gitignore` also prevents secrets from being saved into Git by accident.

NOT YET DONE

Self-signup is currently open (fine for development, review before launch). The app's local phone database is not encrypted beyond what the phone itself provides.

10. The files

You do not need to read code, but it helps to know where things live.

```
D:\Project_Alpha_X
|-- ZUNFT_BUILD_PLAN.md          the original plan and product brief
|-- README.md                    how to start, how to run tests
|-- .env.example                 which secret settings exist (no real secrets)
|-- docs\                        this guide
|
|-- supabase\                    THE CLOUD
|   |-- config.toml              local Supabase settings
|   |-- seed.sql                 demo data for a local test database
|   |-- migrations\              database changes, in order
|   |     20261001...init_zunft_schema.sql     the original tables
|   |     20261002...harden_schema.sql         locks, numbering, storage rules
|   |     20261003...capture_idempotency.sql   duplicate-note protection
|   |     20261004...document_workflow.sql     create_document(), status workflow
|   |-- functions\
|   |     process-capture\index.ts             the "head chef"
|   |     _shared\draft.ts, draft_test.ts      AI-answer checker + totals (+ tests)
|   |-- tests\schema.test.mjs                  33 database tests
|
|-- zunft_app\                   THE PHONE APP
    |-- android\ ios\ windows\   per-platform wrappers
    |-- assets\fonts\           Lato font for the PDF (open licence)
    |-- lib\
    |   |-- main.dart            app start
    |   |-- app.dart             the three-tab main screen
    |   |-- core\
    |   |   |-- config\env.dart            cloud address settings
    |   |   |-- database\app_database.dart the phone's notebook + outbox
    |   |   |-- sync\                      the "filing clerk" (engine, remote, controller)
    |   |   |-- theme\ utils\ constants\   colours, price parsing, validators
    |   |-- features\
    |       |-- auth\            login, onboarding, business details
    |       |-- customers\       customer list + edit form
    |       |-- jobs\            job list, job screen with its notes, edit form
    |       |-- data\            repository + live data providers + sync tile
    |       |-- capture\         job picker, recording sheet, note queue, processor
    |       |-- review\          draft review screen
    |       |-- documents\      finalize service, PDF, Belege tab
    |-- test\                    264 app tests
```

11. Testing - how we know it works

A **test** is a little program that uses another program and checks the result. If we change something later and break it, tests shout immediately. **351 tests** currently pass.

Where	Count	What they check
App (Flutter)	264	Price formatting and parsing; tax split and totals; PDF is created; form checks (IBAN, VAT ID, e-mail, rates); login, registration and onboarding screens; business details editing; sync engine (push order, offline, rejected changes, deletes, paging, lost-edit protection); customers and jobs screens; the sync tile; the note queue (upload, retries, offline, duplicates, vanished files); the recording sheet (permission, too short, 3-minute limit, closing mid-recording, typed notes); the list of waiting notes; the duplicate-customer check; the team screen, invitations, joining with a code, assigned jobs, what an employee sees, switched-off access; removing people, leaving, deleting an account, the privacy notice, the phone forgetting a business; the job picker and the job screen (notes of several workers, removing a note, combining notes into one invoice, a job without customer); the PDF (fonts, all variants, several pages); the finalize service (numbers, retries, offline, rejections, housekeeping, status, sync of documents); the review and finalize screens; the Belege tab.
Cloud function (Deno)	11	Totals match the app; flat fee counts as labour; the AI's mistakes are caught (bad types, negative numbers, a customer ID in the AI's answer is ignored; a spoken name is only a hint); a repeated request gets the stored result back.
Database (SQL)	76	Businesses cannot see each other; cross-links refused; role protection; audit log cannot be edited; invoice numbers sequential, per business, gap-free after a failed save; locked invoices frozen; storage folder rules; the same note id cannot be logged twice in one business; create_document (totals computed by the cloud, retries, every invalid or missing field rejected, nothing stored and no number used up on rejection, the 250 EUR rule, business completeness, foreign references, no direct inserts); frozen content with the status workflow; the insert guard; teams (visibility per role, assignments, invitations, seat limits, switching people off, documents for boss and office only, protected recordings; removing, leaving and deleting; the author name staying on a note); job notes (colleagues see authors, no faked author, removing a note, notes used up exactly once, foreign notes untouched).

WE LOOKED AT THE OUTPUT, TOO

For the PDF I rendered real samples and inspected them as images (Chapter 6, Step 6), because "a valid PDF file" says nothing about how it looks.

WE TESTED THE TESTS

Tests that always pass prove nothing. For the most delicate protection (the lost-edit guard) I switched it off on purpose and confirmed the test then fails, then switched it back on.

What we could not check - important

NOT VERIFIED AGAINST THE REAL THING

- **No real Supabase yet.** This computer has no Docker and no Supabase tool, so the database tests used a look-alike engine, and the sync tests used a pretend cloud. The *logic* is thoroughly tested; the *real connection* (exact web-request details, timestamps) is the first thing to check on first run.
- **Never run on a phone or emulator.** Tests render screens in a simulator with a *pretend microphone*. Whether the real microphone, the permission pop-ups on Android and iPhone, and the audio file format work has not been checked on a device.
- **Gemini and Deepgram never called.** We have no keys yet, so nobody has seen how well the AI really turns German speech into drafts. The prompt wording and the model name (`gemini-2.5-flash-lite`) must be confirmed against Google's current list.
- **Tax rule from memory** (flat fee = labour). Verify before real use.

12. What is still open

Item	Plain explanation	When
Finalizing needs internet	The number comes from the cloud, so a document can only be issued online. Drafts can be prepared offline.	By design
No cancellation flow	You cannot cancel an issued invoice in the app. Legally a correction needs a cancellation invoice (Stornorechnung); today you create a new document instead.	Later
Original PDF vs. recreated PDF	The original PDF is stored online. If it cannot be downloaded, the app recreates it from the stored data using your *current* business details, so it could differ slightly if you changed them since.	Later
E-invoices (XRechnung / ZUGFeRD)	As I understand it, businesses in Germany must be able to *receive* e-invoices since 2025 and must *issue* them to other businesses from 2027 or 2028, depending on turnover. Invoices to private customers are not affected. Not built yet (planned "Phase 4"). Please verify the dates.	Phase 4
Invoice layout review	The invoice follows § 14 UStG as I know it, but I am not a tax adviser. Have someone check one real sample before you send invoices to customers.	Before launch
Reminders (Mahnungen)	The app shows open invoices but sends no reminders.	Later
Data protection (DSGVO)	Voice recordings (which contain whatever names are spoken) are sent to outside companies (Deepgram, Google). Your customer list is no longer sent. Before real customers use it you need data-processing agreements with them, a privacy policy, and ideally EU data regions.	Before launch
Recording lost if the app dies	A note is only queued when you tap stop. If the phone crashes or the battery dies *during* a recording, that recording is lost.	Later
Deploying the function	The Edge Function and the eight migrations must be deployed to a real Supabase project, with the secret keys set there.	First real test
Payments for plans	Seats exist and are enforced, but the limit is set by us by hand. There is no purchase or upgrade screen, and the app-store rules for selling subscriptions in an app still need checking.	Before selling
Camera scanner	The QR scanner for joining a business could not be tried here (needs a real phone). Typing the code works as a fallback.	First real test
Changing a role	A role is fixed when invited. (One person belongs to one business, by design.)	Later
Merging duplicate customers	New duplicates are warned about, but two customers that already exist cannot be merged into one yet.	Later

No audio playback	The recording stays in the cloud, and the written text can be shown per note, but there is no play button yet.	Later
Edit conflicts	"Last write wins" can overwrite simultaneous edits to the same record.	Later, if needed
No automatic fix for rejected changes	The tile tells you something was rejected but offers no "resolve" screen.	Later
First "save point"	The first commit is made (local only). Pushing to GitHub comes when you ask.	Done locally
Windows Developer Mode	Needed on this PC to run the app locally. Turn it on once via `start ms-settings:developers`.	You

13. What comes next

Step 5 - The real capture flow done

Built (see Chapter 6). The first real test needs the keys and a deployed cloud project, listed below.

Step 6 - Documents done

Built (see Chapter 6).

Step 7 - Quality and release

1. Test the voice-to-draft quality on about 25 real German voice notes (this is where we decide on Deepgram vs. Gemini audio and which Gemini tier).
2. Automatic checks on every change (CI): **written** (`.github/workflows/ci.yml` : analysis plus all three test suites), but it only starts running once the project is on GitHub. It was not run yet.
3. Test on real devices; prepare the store releases.

What you will need to provide

Item	What it is	Needed for
Supabase project	A free account at supabase.com, or Docker on this PC for a local copy.	Running the cloud for real
Gemini API key	From Google AI Studio.	The first real test (now possible)
Deepgram API key	From deepgram.com (new accounts get free credit).	The first real test (now possible)
An Android phone (or emulator)	To tap through the app. This PC also needs Android Studio, *or* you use the GitHub build (Chapter 15, Part C).	All device testing
A Mac with Xcode	Apple only allows iPhone apps to be built on a Mac.	The iPhone version
Developer accounts	Google Play (one-time fee) and Apple Developer (yearly fee).	Publishing
10-25 real German voice notes	Recordings of the kind of thing a craftsman would actually say - ideally from a real electrician.	Judging AI quality
A tax check	When you can: have someone confirm the § 35a handling.	Before real invoices

14. Your questions, answered

You asked these as "unguided observations". Most of them are good ones. Each answer says what is true **today**, and whether it should become a job.

Is Zunft for one tradesman or for small companies?

Both, but today only the "boss" side exists. The database was designed for a business with several people (every person has a profile that points to a business, with a role). But the app currently only has screens for the owner. There is no screen to add employees, and no time tracking.

> **MY RECOMMENDATION**
>
> Do not build a time-tracking system or a desktop program yet. Time tracking is a whole separate product area (existing tools such as Clockodo or Toggl already do it well). First prove the core idea: voice note becomes invoice. Later, the cheap way to give the boss an overview is a web page version of this same app (Flutter can do that) plus a "download as spreadsheet" button that existing time tools can read. One app, not two.

Are we allowed to send customer data to outside companies?

I am not a lawyer, so treat this as a briefing and not as legal advice. As I understand the GDPR (DSGVO):

- **It is allowed** to use outside companies to process data, if you have a signed **data-processing agreement** (*Auftragsverarbeitungsvertrag*, AVV) with each one, your customers can read a **privacy policy**, and any transfer outside the EU has a legal safeguard.
- **Today we use two such companies**: Deepgram (voice to text) and Google (Gemini). Both are US companies. Both offer the paperwork, but you must actually accept it in their dashboards.
- **Free versus paid matters.** As far as I know, Google's *free* Gemini tier may use your data to improve its products. For real customers, use the **paid** tier (billing turned on), where Google says it does not train on your data. Please confirm this in Google's current terms.

Idea	Does it help?
Use **Gemini for the audio too** (it can listen)	Yes: one outside company fewer, and less paperwork. We will compare quality and cost in the voice-note test.
Use **Mistral** (French company)	It can help with "European" paperwork. As I know it, Mistral also offers an audio model (Voxtral). We would test it the same way. It is not automatically better: what matters is the contract, the data region, and quality in German trade language.

Choose an **EU data region** (for example Google Vertex AI in Europe)	Yes, this is probably the strongest single step. It needs a slightly different setup, which I can build once you decide.
Hide personal data from the AI	Partly. **Already done in the redesign:** the AI no longer gets your customer list at all (no names, first names or cities from the database), and it never chooses a customer. What still goes out is what is *spoken*: often a name, sometimes an address. We never send street addresses, phone numbers, bank details or e-mail from your database. We could go further and blank out spoken names before the text reaches the AI, but the voice itself contains names, so the speech-to-text company always hears them. We can lower the amount of personal data, not make it zero.

WHAT TO DO

1) Use the paid Gemini tier. 2) Accept the data-processing agreement of each provider. 3) After the voice test, decide on one provider and an EU region. 4) Write a privacy policy (a lawyer or a template service). This is a launch requirement, not a blocker for testing with your own fake data.

Newer Gemini models (3.6, 3.8 Flash)?

I cannot confirm those model names from here: my knowledge may be older than yours. That is exactly why the model name is a **setting** (`LLM_MODEL`) and not written into the code. You change one secret and deploy again; no code changes. In Google AI Studio, open the model list, copy the exact ID of the newest Flash model, and put it in. We will pick the final model with the voice-note test, using quality per euro as the measure.

How does a new employee get an account?

Built (see Step 8). The boss shows a QR code and a short code; the employee registers with their own e-mail and joins with it. One rule I kept: the boss never shares their own login. Every person gets their own account so that you can see who did what, and so that you can switch off one person (a leaving employee) without changing everyone's password.

1. Boss opens "Team" and types the employee's e-mail

↓

2. The cloud sends an invitation e-mail

Done by a server function, because creating accounts needs the secret key, which never goes in the app.

↓

3. Employee taps the link, chooses a password, and lands inside the boss's business

With the role "employee": can record and see jobs, cannot change business settings.

Decision for you later: may employees see *all* invoices and customers, or only their own jobs? Today everyone in a business would see everything.

Fingerprint and face login

Yes, easy and worth doing. It does not replace the account. The phone keeps you logged in, and fingerprint/face is simply a lock that opens the app (Android and iPhone both support it). It needs a real phone to test properly, so I want to add it right after the first real test and not before.

Worker goes from job to job with the internet off

This is what the queue was built for. Every recording gets its own unique number and is saved on the phone. When the internet returns, the app uploads them one by one, in order. A failed upload retries three times, and a real rejection is shown clearly and nothing silently disappears. Customers and jobs typed offline sync the same way.

Honest limits: (1) **issuing** the final numbered invoice needs internet; (2) a recording is only queued when you tap stop, so a phone crash *during* a recording loses it; (3) I have tested all this with simulated networks, not with a real phone in a real basement. That is what the first real test is for.

"Each worker has their own customers, so two people can't collide"

Small correction: customers belong to the business, not to a worker. Two employees see the same customer list. Two good things: recordings can never collide (each phone has its own queue and every note is stored on the job it was recorded for). Two weak spots: two people may create the same customer twice (the app now warns when name and address, phone or e-mail match, but not for same-name people at different addresses), or edit the same customer at the same time (the last save wins). A "merge duplicates" function is a sensible later feature.

Is the voice recording kept, for checking the AI?

Partly. The text the AI heard (the transcript) is saved with every job, and the audio file stays in the cloud storage. Only the copy on the phone is deleted once the cloud has it. **Done:** every note on the job screen has "Was wurde gesagt?", which shows the written text, so the craftsman can check what the AI understood. **Missing:** a "play recording" button. I have put it on the list. (Under the GDPR you should also decide how long audio is kept; I suggest deleting it after a set time, for example 90 days.)

Do 10 notes for Herr Müller all land on one invoice?

They do now - on purpose, and only on the right job. This was your best observation, and it has been built (see Step 6b). In the first version every note was its own draft and the AI guessed the customer. Now:

- You **choose the job first**; the job knows its customer. A big banner in the recording sheet always shows "AUFNAHME FÜR: customer / job", with "Ändern" if you picked wrong.

- The AI never picks a customer, so "Frau Müller" can no longer turn out to be the wrong Müller.
- All notes of a job - from any worker, on any day - collect on that job. One button creates **one** invoice from all of them. A multi-day job (Schiller on Monday, Lapke on Tuesday, Schiller again on Wednesday) is simply two jobs: you record into whichever one you are at.
- A note can be removed from the job (bin icon) if it is wrong.

And to your earlier thought "each worker has their own copy": invoices belong to the **company** (one number sequence, one invoice per job). What each worker keeps is their *name on every note*, so you can see who was there and when.

My recommended order

#	What	Why this order
1	**First real test** (Chapter 15)	Everything so far is only tested with simulations. Real problems will show up here.
2	Voice-note quality test, then choose provider and EU region	Settles Deepgram vs. Gemini vs. Mistral and the privacy design in one go.
3	Job-linked recording and combined notes **(done, Step 6b)**, duplicate-customer warning **(done)**, team, invitations, assigned jobs, seats **(done, Step 8)**. Next: play button, merge duplicates	The big gaps you spotted are fixed; these are the small ones.
4	Team invites and roles	Needed as soon as a second person uses it.
5	Fingerprint / face lock	Small, best tested on a real phone.
6	Boss overview, time entries, web version	Only after the basics are loved by a real customer.
6b	**Job scheduler and notifications.** The boss gives a job one or more *appointments* (date, start time, how long, who). The worker gets a push message and sees "Today / This week" with the appointments. Self-employed people use the same calendar for their own jobs. Optional: export to the phone's calendar, remind the customer.	Good fit: assignment already exists, so this adds *when* to it. Needs a separate appointments list (a job can have several visits, e.g. Monday to Friday and again on Wednesday), push notifications (Firebase for Android; Apple's service, which needs the paid Apple developer account, for iPhone) and conflict warnings (same person twice at the same time). Keep it a simple calendar; no route planning or automatic dispatch.
7	**Several languages** (app screens and the customer's invoice language), so workers who do not read German well can use it, and the app can be sold outside Germany	Planned. The app's texts are German only today; they have to be moved into translation files first, and the invoice rules (DIN 5008, § 35a) stay German-law specific.

15. Your next steps: a simple checklist

GOOD NEWS

You do not need to understand any of this. Do the steps in order, tick them off, and tell me when you are stuck or done. Anything that goes wrong: copy the red text and send it to me. Developer Mode is already on, so that part is done.

Part A - Create three free accounts (about 20 minutes)

1. **Supabase** (the cloud database). Go to `supabase.com`, sign up, click **New project**. Name: `zunft`. Choose a **region in Europe (Frankfurt)**. Write down the **database password** you choose. Wait about 2 minutes until it is ready.
2. **Collect 3 values from Supabase.** (a) The *publishable key*: Project Settings → API Keys. The *Project URL* is not shown there; build it yourself from the reference: `https://XXXX.supabase.co` (the address in your browser is the dashboard page, which is the wrong one). (b) *Project reference*: the short text in the URL, `https://XXXX.supabase.co`, that is the XXXX. Keep them in a text file. **Never** send me the "service_role" or "secret" key.
3. **Gemini key.** Go to `aistudio.google.com` → **Get API key**. For the test, the free tier is fine with fake data. Later, switch on billing before using real customer data.
4. **Deepgram key.** Sign up at `deepgram.com` (free starting credit) and create an API key.

Part B - Put the cloud part online (about 15 minutes)

Open a terminal in the folder `D:\Project_Alpha_X` and run these one at a time. (In Claude Code you can type `!` before a command to run it. The first command may ask to install a small helper: say yes.)

5. `npx supabase login` — a browser opens; allow it.
6. `npx supabase link --project-ref XXXX` — use your project reference. It asks for the database password from step 1.
7. `npx supabase db push` — builds the tables (eight migrations). Answer **Y**.

IF THIS STEP SHOWS AN ERROR

Send me the exact text. The first real run found one problem that all my automatic tests could not (the hosted database keeps some add-ons in a separate area my first file could not see). It was fixed by not needing those add-ons at all, and the eighth migration closes a related gap in who may call which cloud function. If an earlier attempt stopped with an error, simply run the same command again: nothing half-finished is left behind.

8. `npx supabase secrets set GEMINI_API_KEY=your-key DEEPGRAM_API_KEY=your-key LLM_MODEL=gemini-2.5-flash-lite` — the secrets stay in the cloud; nobody sees them.
9. `npx supabase functions deploy process-capture` — puts the AI function online.

> **IF SUPABASE ASKS FOR DOCKER**
> It is only needed for a local copy. For the online project you do not need it. Tell me what it said and I will adjust.

Part C - Get the app onto a phone

> **FOUND OUT THE HARD WAY: THIS PC CANNOT BUILD ANDROID APPS YET**
> I checked with Flutter's own health check (`flutter doctor`): the **Android SDK** (the toolbox Google provides for building Android apps) is not installed here. So the command in the old checklist would have failed. Pick one of the two ways below.

WAY 1 - INSTALL ANDROID STUDIO ON THIS PC (ABOUT 45 MINUTES, ABOUT 10 GB)

10. Download **Android Studio** from `developer.android.com/studio` , install it, start it once and let it download the "SDK" when it offers. Accept the licences. Then in a terminal: `flutter doctor --android-licenses` (press **y** every time) and `flutter doctor` again: "Android toolchain" should now have a green tick.
11. **Android phone:** Settings → About phone → tap *Build number* 7 times → back → Developer options → turn on **USB debugging**. Plug it in with a cable and accept the prompt on the phone. In the terminal: `cd zunft_app` , then `flutter devices` : your phone should be listed. (No phone? In Android Studio create a virtual phone under Device Manager and start it.)
12. `flutter run --dart-define=SUPABASE_URL=https://XXXX.supabase.co --dart-define=SUPABASE_ANON_KEY=your-publishable-key` — the first build takes several minutes. That is normal.

WAY 2 - LET GITHUB BUILD THE APP FOR YOU (NO ANDROID STUDIO; NEEDS A PHONE)

I added an automatic build recipe to the project (`.github/workflows/ci.yml`). If you put the project on GitHub, GitHub's computers build the Android app and hand you a file to install on your phone. I could not test this recipe from here, so expect that it may need one small fix.

10. Create a **private** repository on `github.com` and push the project (tell me when it exists and I will help; or, if you have the GitHub CLI, `gh repo create zunft --private --source . --push`).
11. On GitHub: **Settings** → **Secrets and variables** → **Actions** → **Variables tab** → **New repository variable**. Make two: `SUPABASE_URL` (the project URL) and `SUPABASE_ANON_KEY` (the *publishable* key). These two are meant to be public. **Do not** put the Gemini or Deepgram key here.

12. Open the **Actions** tab → **CI** → **Run workflow**. After about 10 minutes, open the finished run and download **zunft-android-apk**. Unzip it, copy `app-release.apk` to your phone (cable, e-mail to yourself, or Drive), tap it, and allow "install from this source". The app then starts like any other.

Bonus: the same recipe also runs all the automatic tests on every change, so a mistake is noticed within minutes.

Part D - The real test (about 40 minutes)

13. In the app: register, then fill in the business details (invent them or use real ones; use a **fake** tax number for now).

14. **Customer and job.** In the **Kunden** tab add a customer (for example "Thomas Müller, Hauptstraße 5, 20095 Hamburg"). In the **Aufträge** tab tap **Neuer Auftrag**, title "Test Steckdose Keller", choose that customer, save.

15. **First note.** Tap the job, then **Notiz zu diesem Auftrag aufnehmen**, then the microphone, and say, in German: "*Steckdose im Keller gesetzt, zwei Stunden Arbeit, Material etwa vierzig Euro.*" (You do not need to say the customer's name: the job already has it.) Stop. After a few seconds the note appears on the job with its positions.

16. **Second note.** Record another one on the same job, for example "*Anfahrt pauschal fünfunddreißig Euro, eine Stunde Fehlersuche*". Check that both notes show up, and that "Was wurde gesagt?" shows what the app understood.

17. **One invoice.** Tap **Alle Notizen prüfen und abrechnen**. Check the customer, the labour/material split and the sums of both notes together. Press **Vorschau als PDF**, then **Rechnung fertigstellen** and look at the PDF. Go back to the job: the notes should be gone (they were used up).

18. **Airplane-mode test:** create a second job for another customer. Turn on airplane mode, record one note on each job, then switch the internet back on. Both notes should reach their own job on their own.

19. Send me: what worked, what did not, any red error text, and a photo of the PDF if it looks wrong. Then record **10 to 25 more** voice notes the way a real craftsman talks (ideally have a real one record some) and keep the audio files; they are the test set for choosing the AI.

Part E - Optional: test the team features (needs a second e-mail and, ideally, a second phone)

Do this only after Part D worked. A new business has **one seat** and only we can raise it, so you raise it by hand:

20. In Supabase open **SQL Editor** → **New query**, paste `update businesses set seat_limit = 5;` and press **Run**. (This is a test shortcut: it is exactly what we will later do automatically when someone buys a plan. Run it only while you have a single test business.)

21. In the app, signed in as the owner: **Betrieb** → **Team** → **Mitarbeiter einladen** → Geselle. A QR code and an 8-character code appear.

22 On the second phone (or after signing out on the first): install the app, **register with a second e-mail**, tap **Mit Einladungscode beitreten**, enter a name and scan the QR code from the first phone (or type the code). You land in the app with only Aufträge, Kunden and Betrieb: no Belege.

23 As the owner: open a job → **Zuweisen** → tick the new person → Speichern. On the employee's phone the job appears after the next sync (close and reopen the app, or wait a few minutes). The employee can record a note on it, but cannot bill and sees no other job.

24 As the owner: **Team** → **Abschalten** for that person. The employee's app now shows "Dein Zugang wurde abgeschaltet". Please tell me whether the **camera scanner** worked.

25 As the owner: **Team** → **the red person-with-minus icon** (remove) for that person. On their phone, after a few minutes (or after reopening the app), they should land on "Betrieb einrichten": their account still works, and they could start their own business. Check that the old jobs are gone from their phone.

STUCK? THAT IS NORMAL

The first run of any app against a real cloud almost always has one or two surprises (a wrong key, a missing setting). Send me the exact message and we fix it together. You cannot break anything permanent: the project is empty.

PARALLEL, NO RUSH

Look into: a data-processing agreement with each provider (Chapter 14), a privacy policy, someone to check the invoice layout and the § 35a handling, and an Apple Developer account plus a Mac if you want the iPhone version.

SUMMARY IN ONE PARAGRAPH

Zunft is a phone app that turns a spoken job note into a legally usable German invoice. So far we have made the cloud side safe, built login and business setup, a local database with offline sync, the recording queue that turns a note into a draft even with bad reception, and now the last stretch: checking the draft, issuing a numbered and locked document, and a proper PDF - then a redesign so that a job collects all notes of all workers into one invoice - all verified by 351 automatic tests. Still ahead: a first real test with real keys and phones, a review of the invoice layout, then release (Step 7).